

10-bit SAR ADC Design in Sky130 – Project Report

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1 Introduction and Specifications

The successive-approximation (SAR) ADC is the most commonly deployed topology of analog-to-digital converter. Owing to its low power consumption, low latency and compact form factor, it has proven supremely versatile, with notable usage in IoT devices, medical equipment and microcontrollers. This document describes an original 10-bit SAR ADC designed and verified in the SkyWater Sky130 open-source 130nm process — a CMOS PDK developed by SkyWater Technology and Google, providing full device model access without NDA — targeting the specifications in Table 1. The design covers the complete front-end flow: behavioral modeling from first principles, transistor-level schematic capture in xschem, ngspice simulation against Sky130 device models, and Verilog RTL synthesized to Sky130 standard cells via Yosys. Physical layout is out of scope; the project targets schematic-level verification.

Table 1: Design Specifications

Parameter	Value
Resolution	10 bits
ENOB	≥ 9 bits
Supply Voltage, V_{DD}	1.8 V
Input Range	0 – 1.8 V
Process	SkyWater Sky130, 130nm

The design followed a bottom-up flow, progressing sequentially through each of the following stages:

1. First-principles sizing and yield analysis via Python behavioral model
2. Block-level schematic capture in xschem and isolation verification in ngspice
3. Verilog RTL synthesis to Sky130 standard cells via Yosys
4. Full-ADC SPICE integration and static performance characterisation

This document reports the design decisions, block-level verification, integration debugging and static characterisation of the 10-bit SAR ADC. Ultimately, the design achieves a static ENOB of 7.27 bits, falling short of its 9-bit target, attributable to a characterised top-plate feedthrough mechanism, detailed in Section ??.